

Finite-Prefix Corrections and Dual Certificates for Polyomino Growth

An exact bound $\lambda \leq 4.498$
and a certificate theorem for convolution recurrences

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Abstract

We develop two extensions of the seventeen-neighborhood polyomino project in ProveIt. First, exact small-size occurrence counts are incorporated into a positive convolution system by subtracting its finite coefficient defect. A nonnegative tail formulation makes this correction rigorous without assuming convergence of any generating function. Enumerating the seventeen marked neighborhoods through size 18 gives an exact rational certificate for

$$\lambda \leq \frac{2249}{500} = 4.498 < 4.5,$$

improving the repository's 4.5235 upper bound. Second, balanced integer weights on monomials produce finite arithmetic certificates of divergence. We prove completeness for productive, strongly connected, nonlinear positive polynomial systems, thereby providing a certificate-based answer to Bui's Question 2 for this class. An explicit 53-weight certificate and the repository's primal certificate bracket the uncorrected recurrence growth between 4.52349 and 4.5235. We also prove a spectral stability criterion for the prefix hierarchy and give a counterexample in which arbitrarily long exact prefixes still fail to recover the true exponential growth. Full proofs, exact data, enumerators, rational verifiers, and proposed research directions are supplied.

Status and scope. The new bound is a computer-assisted mathematical result, not a newly kernel-checked Lean theorem. The new arguments are proved in this report; the finite enumeration and arithmetic are reproducible. The geometric recurrence lemma is imported from Bui and the pinned repository. The work has not been independently refereed. General geometric-programming duality and polyomino enumeration algorithms are prior tools, not claimed inventions of this report.

[write] In ProveIt the geometric recurrence lemma is more than imported: Lean proves Bui's Lemma 4 recurrences for the actual marked counts (`geometricPublishedBuiRecurrences`), and the project's Lean-verified bound is $\lambda \leq 9047/2000 = 4.5235$. The bound $\lambda \leq 2249/500 = 4.498$ of this report is computer-assisted and is *not* formalized. See [section 1.5](#).

Contents

1	Problem selection, sources, and main contributions	1
1.1	Why this repository thread	1
1.2	What is proved here	1
1.3	Novelty and dependency boundaries	2
1.4	[write] Notation conventions	2
1.5	[write] What ProveIt has formalized, and what it has not	3
2	Polyomino counts and the positive polynomial system	4
2.1	Counting conventions	4
2.2	Exact offset specification	5
2.3	The imported coefficient inequalities	6
3	Exact-prefix correction: the general theorem	6
3.1	Proper positive polynomial systems	7
3.2	The finite defect and the nonnegative tail map	7
3.3	Monotonicity of the hierarchy	9
4	An exact upper bound below 4.5	9
4.1	The certificate	9
4.2	Discovery is not verification	10
4.3	A reproducible finite enumeration	11
4.4	Finite-machine safety and independent checks	12
5	Balanced-monomial certificates of divergence	13
5.1	The certificate format	13
5.2	From nonexistence to a lower growth bound	14
6	Completeness of the certificate alternative	14
6.1	The hypotheses	14
6.2	Compactness of bounded multiplicative residuals	15
6.3	An elementary convex proof	15
6.4	What this settles, and what it does not	16
7	A compact dual certificate for the original recurrence	16
7.1	The integer weights	17
7.2	Exact verification without enormous integer products	18
7.3	How the weights were discovered	18
8	The limits of the exact-prefix hierarchy	19
8.1	A spectral criterion	19
8.2	An exact counterexample to asymptotic exactness	20
8.3	A conditional sensitivity formula for choosing improvements	21
9	Verification, formalization, and reproducibility	21
9.1	The proof dependency chain	21
9.2	Archive contents and commands	22
9.3	A focused Lean extension plan	22
10	Proposed research questions and next projects	23
11	Conclusion	24
A	Complete marked profiles through size eighteen	25
B	A minimal exact certificate checker	26
C	Source audit and claim classification	26

D [write] Provenance, intake verification and non-claims	27
D.1 Provenance	27
D.2 Intake verification	27
D.3 What this report does not claim	28

1 Problem selection, sources, and main contributions

1.1 Why this repository thread

The ProveIt repository contains a concrete project on fixed square-lattice polyominoes at

Combinatorics/Polyominoes/KlarnerConstant/.

Its existing endpoint is an upper bound of $9047/2000 = 4.5235$ for Klarner’s constant. The relevant sources were read at commit

e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

In particular, the project README, the independent rational verifier, `Patterns.lean`, and `GeometricComplete.lean` fix the mathematical and computational interface used here [2, 3, 4, 5, 6]. The last of these files assembles the seventeen geometric recurrences for actual marked polyomino counts and then applies the 4.5235 certificate. We inspected those source statements; we did not run a fresh Lean build.

The published input is Vuong Bui’s *A convolutional approach to bounding the number of polyominoes*, arXiv:2511.00461v2, dated 6 May 2026 [1]. Its Lemma 4 and Appendix B give the seventeen-neighborhood recurrence system. Its Section 3, Question 2 asks for compact, readily verifiable lower-growth certificates for convolution recurrences. This is a particularly suitable target: the geometric inequalities are already available, the numerical improvement can have an exact proof boundary, and the certificate question has a precise published formulation.

[write] Earlier upper bounds. The delivered text compares its bound only with Bui’s 4.5238 and the repository’s 4.5235. For context, the repository’s research note on the 4.5235 bound [7] records the upper-bound history as follows: Eden obtained 6.75 [8]; Klarner and Rivest successively obtained 5, 4.83 and 4.649551, the last by the procedure of [9]; Barequet and Shalah reached 4.5252 using over twenty-three trillion local configurations [10]; Bui’s seventeen neighborhood types then gave 4.5238 [1], and the repository’s exact supersolution for Bui’s unchanged system gave $9047/2000 = 4.5235$, which is kernel-checked in Lean (section 1.5). The bound 4.498 of theorem 4.1 continues this list, but it is computer-assisted and not formalized. None of the earlier methods used exact marked prefixes of the kind introduced here; this is a statement about the cited sources, not a priority search.

1.2 What is proved here

The first result concerns actual polyominoes. If A_n is the number of translation classes of n -cell polyominoes, then the data and certificate in this report give

$$A_n \leq \frac{927884613}{10^9} \left(\frac{2249}{500} \right)^n \quad (n \geq 1), \quad \lambda \leq \frac{2249}{500}. \quad (1)$$

The factor $927884613/10^9$ is the G coordinate of the supplied rational supersolution. No numerical eigenvalue or floating-point root appears in the verification of (1).

The second result concerns the equality recurrences used as majorants. We give a theorem of alternatives: under explicit productivity, irreducibility, and nonlinearity hypotheses, either a positive supersolution exists or a balanced integer-monomial certificate disproves its existence. The two alternatives cannot occur together. This supplies the requested type of lower-growth certificate for a class containing Bui’s seventeen-variable system. The explicit example proves

$$4.52349 \leq \mu_{\text{original}} \leq 4.5235. \quad (2)$$

Here μ_{original} is the exponential growth of the *uncorrected equality recurrence*, not the polyomino growth constant λ . In particular, the lower inequality in (2) is *not* a lower bound on λ .

The third result describes both the power and a limitation of exact-prefix corrections. The corrected majorants decrease coefficientwise as more exact terms are supplied. They eventually admit certificates at any evaluation point where the linearized system at the true series has spectral radius less than one. Conversely, a supercritical linearization obstructs the whole hierarchy under a mild nonzero-tail condition. A scalar example has true growth 1 but corrected-majorant growth tending to 3.

1.3 Novelty and dependency boundaries

The proposed research contributions are the explicit 4.498 polyomino bound, the finite-prefix application and its exact data, the arithmetic response to Bui’s certificate question, and the accompanying analysis of the prefix hierarchy. The examined repository snapshot and primary sources did not contain this 4.498 certificate or this marked-prefix calculation. That is a limited novelty assessment, not an exhaustive priority determination.

Several ingredients are established mathematics. The geometric inequalities come from Bui. Enumeration by canonical include/exclude growth is in the tradition of Redelmeier [11]. Positive monomial inequalities, logarithmic convexification, and entropy-like dual expressions belong to geometric programming [13, 14, 15]. We do not present those tools as new discoveries. The proof of the certificate alternative is included to make its hypotheses, strictness, and rational verification completely explicit in the recurrence setting.

Three distinct levels. The geometric recurrence theorem is an imported mathematical input. The new prefix and dual-certificate theorems have written proofs here. The headline numerical bound additionally uses the finite enumeration in Appendix A, whose source programs and exact outputs are supplied. Arithmetic verification is not a substitute for verifying enumeration.

[write] In ProveIt the first level is stronger than “imported”: it is a kernel-checked Lean theorem about the actual counts (section 1.5). The second and third levels are not formalized.

1.4 [write] Notation conventions

This subsection was added when the manuscript was written into ProveIt. No delivered symbol was renamed and no normalization was changed; the conventions below only fix how the delivered symbols are to be read.

Coordinate letters. The seventeen upper-case letters of (4) name the marked neighborhood types and their counting sequences ($C_n, \dots, Z_n$, also written $a_{i,n}$ with $i \in \mathcal{I}$); the matching lower-case letters $c, \dots, z$ are the coordinates of the vector $\mathbf{X}$ in (6) and the row names of table 4. A coordinate is always a bare letter or a subscript index ($v_G, a_{D,7}, \Phi_P$). Many of these letters are reused for other objects, which are recognized by a subscript N , by calligraphic type, or by the local context listed here:

Letter	Other meanings in this report
P	$P_N = [\xi^{\leq N}]a$, the exact prefix; a polyomino P (section 2 and proposition 4.2)
D	D_N , the finite defect; the defect polynomial D in (24)
T, Y	$T_N = [\xi^{\leq N}]\Phi(\xi, P_N)$; $Y_N = B_N - P_N$
E	forbidden offset set; the enumerator's excluded set; the defect improvement in (24)
R, S	required offset set, radii R_i ; the enumerator's occupied set, the sets S_K of lemma 6.2
U	the enumerator's frontier; $U = a - P_N$ and a generic subsolution (lemma 3.2 and theorem 3.3)
H	the nonlinear part in $\Phi = LX + H$; the half-plane $\mathcal{H}$
B	the offset sets B, B_4 ; the majorants B_N ; a least fixed point B (section 5)
C	the dual product C ; a constant C (theorem 8.1); the normalizer $C = \sum_i \ell_i \tau_i$
h, q	$h(y)$ of (19), $h = (I - J)^{-1}\mathbf{1}$, a row index; the argument of $\log q$, a degree $q \geq 2$
$v, w, y, p, t, u, x, e,$	certificate vector $v, w = v - P_N(\zeta)$, log-coordinates $y, p_N = P_N(\zeta)$, reals $t, u,$
c, g, r	scalar unknown x , unit vectors e_i , coefficients c_{ik} , a nearest point g , row sums r_i
b	b_n (proof of lemma 3.2); exponents b_{ik} in (15); forcing vector b_N (section 8)

Collisions with the ProveIt project. The project's research note [7] and its Lean files use the same λ , the same $A(n) = A_n$, the same seventeen upper-case sequence names with lower-case coordinates, and the same evaluation point ζ ; its numeric map Φ_ζ is $\Phi(\zeta, \cdot)$ here. Two symbols differ:

- The project writes $b_N = (c_N, \dots, z_N)$ for the ζ -weighted prefix profile and $S_N = \sum_{n \leq N} S(n)\zeta^n$ for one sequence. That profile is $p_N = P_N(\zeta)$ here. *Tempting false reading:* the b_N of section 8 is the project's b_N . It is not; it is the high-degree forcing vector $([\xi^{>N}]\Phi(\xi, P_N))(\zeta)$.
- The project writes a for its rational supersolution; that is v here, while $a = a(\zeta)$ in section 8.

The symbol μ (the growth of an equality recurrence, μ_{original} for Bui's uncorrected system) occurs only in this report. *Tempting false reading:* $\mu_{\text{original}} \geq 4.52349$ is a lower bound on λ . It is not (section 5). The convention $A_0 = 0$ of section 2 differs from the value $a(0) = 1$ of OEIS A001168; positive-size terms agree.

1.5 [write] What ProveIt has formalized, and what it has not

This report belongs to ProveIt's research-report collection and continues the formal project Combinatorics/Polyominoes/KlarnerConstant/; being filed next to, or citing, a Lean development confers no formal status on it. That project proves the following in Lean (namespace LeanProofs.KlarnerConstant; files under Combinatorics/Polyominoes/KlarnerConstant/Lean/KlarnerConstant/, cited below by file name), with no sorry, project axiom or native_decide; its assumption audit is Audit.lean. Line numbers are those of the pin; the project's files are unchanged between the pin and the placement commit.

- *Bui's Lemma 4 for the actual counts.* geometricPublishedBuiRecurrences (GeometricComplete.lean:63) proves PublishedBuiRecurrences for geometricCoefficientProfile: the seventeen recurrences for $n \geq 2$, the published degree-one values, nonnegativity and vanishing at zero, for the finite marked-occurrence counts of the patterns in Patterns.lean. Coefficient by coefficient this is the inequality (7), which the delivered text calls imported; reading the Lean structure as (7) in formal-series notation is not itself a formal step.
- *The five partitions (8) as equalities:* buiF_occurrenceCount_eq_g_add_p, buiG_occurrenceCount_eq_e_add_q, buiH_occurrenceCount_eq_d_add_s, buiR_occurrenceCount_eq_y_add_w and buiT_occurrenceCount_eq_x_add_v (GeometricLinear.lean:279-312).

- *G-domination* $A_n \leq G_n$: `fixedPolyominoCount_le_geometricCoefficientProfile_g` (`GeometricProfile.lean:371`). The other half $G_n \leq nA_n$ of (5) is not formalized; the upper bound does not need it.
- *Supermultiplicativity and Fekete* (3): `fixedPolyominoCount_supermultiplicative` and `tendsto_realNthRoot_fixedPolyominoCount_growthSup` (`GeometricComplete.lean:97`).
- *The Lean-verified bound* $\lambda \leq 9047/2000 = 4.5235$: `certificate_isSupersolution` (`Certificate.lean:232`), `fixedPolyominoCount_le_9047_div_2000_pow` (`GeometricComplete.lean:69`), its real cast (`:76`) and `growthSup_fixedPolyominoCount_le_9047_div_2000` (`:88`).

The one human boundary of that development is the transcription of Bui’s page-15 diagrams into the offset tables of `Patterns.lean`.

Nothing new in this report is formalized: not [lemma 3.2](#), [theorems 3.3, 3.4](#) and [3.6](#), not the size- ≤ 18 table, not [theorem 4.1](#) ($\lambda \leq 2249/500 = 4.498$, which is computer-assisted), and none of [sections 5](#) to [8](#). The project’s endpoint theorems cannot simply be re-instantiated: they are hard-wired to $\zeta = 2000/9047$ (`certificateZeta`) in `PrefixRecurrence.g_le_certificate` and `g_lt_one` (`Recurrence.lean:321, 327`), `CoefficientProfile.g_lt_9047_div_2000_pow` (`:457`) and `dominatedCoefficient_le_9047_div_2000_pow` (`:477`). Only `buiIterate_le_supersolution` (`:283`) and `PrefixRecurrence.le_supersolution` (`:309`) are generic in ζ . A formal 4.498 bound therefore needs, beyond the plan of [section 9](#), a prefix-corrected variant of that weighted-prefix engine and a computable bridge to the noncomputable `BuiNeighborhood.occurrenceCount` (`Patterns.lean:140`). As a formalization target, not a claim: the certificates for $N = 7, \dots, 10$ in [table 3](#) (4.5233 down to 4.5194) already improve on 4.5235 and need exact marked counts only through size 7 to 10.

The trust boundary of [theorem 4.1](#) is thus: Lean-proved recurrences (relative to `Patterns.lean`), plus an exhaustive enumeration that is not kernel-checked (regenerated at intake, [section D](#)), plus exactly replayed rational arithmetic (Python, not Lean), plus the written and unrefereed proofs of [section 3](#).

2 Polyomino counts and the positive polynomial system

2.1 Counting conventions

A polyomino is a nonempty finite edge-connected subset of $\mathbb{Z}^2$. Two such subsets are equivalent precisely when one is a translate of the other. Rotations and reflections are not identified. Holes are allowed. Let A_n be the number of equivalence classes with n cells, for $n \geq 1$. For the recurrence calculation we put $A_0 = 0$ and all marked size-zero counts equal to zero. This convention differs from the optional empty-object term $A_0 = 1$ in some sequence tables; positive-size terms are unchanged.

The standard concatenation argument gives $A_{m+n} \geq A_m A_n$ for positive m, n . Fekete’s lemma then gives

$$\lambda = \lim_{n \rightarrow \infty} A_n^{1/n} = \sup_{n \geq 1} A_n^{1/n}. \quad (3)$$

These are background inputs also implemented in the repository’s `Concatenation.lean` and `Asymptotic.lean` [[1, 3, 6](#)].

A neighborhood pattern consists of a required offset set R containing $(0,0)$ and a disjoint forbidden offset set E . An occurrence at a cell a of a polyomino P means

$$a + R \subseteq P, \quad (a + E) \cap P = \emptyset.$$

The marked count for a pattern is the sum of its occurrences over all fixed polyominoes of the given size. Counting occurrences, rather than merely polyominoes which have an occurrence, is essential.

Write $C_n, D_n, \dots, Z_n$ for the seventeen counts, in the coordinate order

$$\mathcal{I} = (C, D, E, F, G, H, P, Q, R, S, T, U, V, W, X, Y, Z). \quad (4)$$

To avoid confusing a coordinate name with a function, the polynomial map will be denoted by Φ .

2.2 Exact offset specification

Let

$$B = \{(-1, -1), (0, -1), (1, -1)\}, \quad B_4 = B \cup \{(2, -1)\}.$$

In [table 1](#), 1, 2, 3 in the required column mean respectively $\{(0, 0)\}$, $\{(0, 0), (1, 0)\}$, and $\{(0, 0), (1, 0), (2, 0)\}$. A blank lattice position imposes no condition. These coordinate sets agree with the repository's pattern table and Bui's page-15 neighborhood diagrams [\[5, 1\]](#).

Type	Required	Forbidden offsets
C	1	$B \cup \{(-1, 1), (1, 1), (-1, 0), (1, 0)\}$
D	1	$B \cup \{(-1, 1), (-1, 0), (1, 0)\}$
E	1	$B \cup \{(-1, 0), (1, 0)\}$
F	1	B
G	1	$B \cup \{(-1, 0)\}$
H	1	$B \cup \{(-1, 1), (-1, 0)\}$
P	2	$\{(0, -1), (1, -1), (2, -1)\}$
Q	2	$B \cup \{(-1, 0)\}$
R	2	$B_4 \cup \{(-1, 1), (-1, 0)\}$
S	2	$B \cup \{(-1, 1), (-1, 0)\}$
T	2	$B_4 \cup \{(-1, 0)\}$
U	2	B_4
V	3	$B_4 \cup \{(-1, 0)\}$
W	3	$B_4 \cup \{(-1, 1), (-1, 0)\}$
X	2	$B_4 \cup \{(-1, 0), (2, 0)\}$
Y	2	$B_4 \cup \{(-1, 1), (-1, 0), (2, 0)\}$
Z	2	$B_4 \cup \{(-1, 1), (2, 1), (-1, 0), (2, 0)\}$

Table 1: The seventeen marked neighborhood patterns.

Every polyomino has a G occurrence at the leftmost cell of its bottom row. There are at most n possible marked cells. Consequently

$$A_n \leq G_n \leq nA_n. \quad (5)$$

The first six size-one counts equal one; the other eleven equal zero.

2.3 The imported coefficient inequalities

Use the formal variable ξ and coordinate vector $\mathbf{X} = (c, d, e, f, g, h, p, q, r, s, t, u, v, w, x, y, z)$. Define

$$\begin{aligned}
\Phi_C &= \xi + \xi e, & \Phi_D &= \xi + \xi g, & \Phi_E &= \xi + \xi f, \\
\Phi_F &= g + p, & \Phi_G &= e + q, & \Phi_H &= d + s, \\
\Phi_P &= eh + qd + xr + vy + uyz, \\
\Phi_Q &= \xi g + \xi ge + \xi^2 u + \xi^2 t g + \xi^2 ru, \\
\Phi_R &= y + w, \\
\Phi_S &= \xi g + \xi e^2 + \xi^2 t + \xi^2 x g + \xi^2 y u, \\
\Phi_T &= x + v, \\
\Phi_U &= dh + sd + yr + wy + uz^2, \\
\Phi_V &= \xi s + \xi^2 g^2 + \xi^2 te + \xi^2 rt, \\
\Phi_W &= \xi s + \xi^2 e g + \xi^2 x e + \xi^2 y t, \\
\Phi_X &= \xi d + \xi^2 g + \xi^2 u, \\
\Phi_Y &= \xi c + \xi^2 g + \xi^2 t, \\
\Phi_Z &= \xi c + \xi^2 e + \xi^2 x.
\end{aligned} \tag{6}$$

There are 53 monomials, each with coefficient one. The order displayed within each row is also the order of the dual weights in [table 4](#). In the first two displayed lines, each coordinate has its own two-term row.

For the true generating series $\mathbf{a}_i(\xi) = \sum_{n \geq 1} a_{i,n} \xi^n$, Bui’s geometric lemma is the coefficientwise inequality

$$\mathbf{a} \leq \Phi(\xi, \mathbf{a}). \tag{7}$$

All products in this equation are formal Cauchy products. Thus convergence is not part of the statement. Equations with a factor ξ^2 , for example, encode a two-cell shift in the corresponding convolution index.

The geometric proof of (7) uses occupancy partitions, injective leaf deletions, and decompositions into connected marked pieces. We import that lemma rather than silently treating a numerical recurrence fit as a geometric proof. For example, the C row deletes its marked cell, whose only possible neighbor is above it. The P and U rows encode multi-piece decompositions; these are not inferred from the finite table. The repository supplies their separate finite-injection modules [1, 6]. [write] Those modules prove the lemma in Lean for the actual counts (`geometricPublishedBuiRecurrences`); so, within ProveIt, the input (7) is a kernel-checked theorem relative to the offset tables of `Patterns.lean`, not an imported assumption ([section 1.5](#)).

The five same-size partitions are in fact exact:

$$F = G + P, \quad G = E + Q, \quad H = D + S, \quad R = Y + W, \quad T = X + V. \tag{8}$$

For example, a type- G anchor either has no right neighbor, yielding type E , or has a right neighbor, yielding type Q . Similar left/right occupancy splits give the other identities. They will serve as independent consistency checks on the occurrence table. [write] The five identities are Lean theorems in ProveIt (`GeometricLinear.lean:279–312`); the table’s agreement with them is checked by the Python verifier, not by Lean.

3 Exact-prefix correction: the general theorem

3.1 Proper positive polynomial systems

Definition 3.1. A proper positive polynomial system is a map $\Phi \in \mathbb{R}_{\geq 0}[\xi, X_1, \dots, X_d]^d$ such that $\Phi(0, 0) = 0$ and the nonnegative matrix

$$L = \Phi_X(0, 0)$$

is nilpotent. Here and below vector inequalities are coordinatewise.

Nilpotence handles same-size linear dependencies. In (6), the only such dependencies are the five rows in (8); their directed graph has no cycle. All remaining occurrences of an unknown are multiplied either by a positive power of ξ or by another unknown. Thus the system is proper. One possible degree-by-degree evaluation order is

$$C, D, E, Q, S, V, W, X, Y, Z, P, U, G, F, H, T, R.$$

The formal development does not require this particular order.

Lemma 3.2 (Formal fixed points and comparison). *A proper positive polynomial system has a unique fixed point in $\xi\mathbb{R}[[\xi]]^d$. Its coefficients are nonnegative, and the iterates from zero increase coefficientwise and stabilize at each finite degree. If $U \in \xi\mathbb{R}_{\geq 0}[[\xi]]^d$ satisfies $U \leq \Phi(\xi, U)$, then U is coefficientwise at most this fixed point.*

Proof. Write $\Phi = LX + H$. Because the unknown series have zero constant terms, the coefficient of degree n in $H(\xi, X)$ depends only on unknown coefficients of degrees strictly less than n . If these have already been chosen, the coefficient vector b_n in a fixed point must satisfy

$$(I - L)b_n = h_n.$$

For some k , $L^k = 0$, so

$$(I - L)^{-1} = I + L + \dots + L^{k-1} \geq 0.$$

This proves existence, uniqueness, and nonnegativity by induction on n . For a subsolution, the same argument gives $(I - L)u_n \leq h_n(u_1, \dots, u_{n-1})$; multiplication by the nonnegative inverse and induction prove comparison.

The iterates from zero are increasing by monotonicity. If all lower-degree coefficients have stabilized, the remaining degree- n error is multiplied by L at each further iteration, and therefore vanishes after at most k steps. This proves finite-degree stabilization. $\square$

The subsolution conclusion uses formal properness. It would be false for arbitrary numerical subsolutions: $2 \leq 2^2$ does not imply that 2 is below the least nonnegative fixed point of $x \mapsto x^2$.

3.2 The finite defect and the nonnegative tail map

Suppose $a \in \xi\mathbb{R}_{\geq 0}[[\xi]]^d$ is a true counting series satisfying (7). For $N \geq 1$, set

$$P_N = [\xi^{\leq N}]a, \quad T_N = [\xi^{\leq N}]\Phi(\xi, P_N), \quad D_N = T_N - P_N. \quad (9)$$

The coefficient of $\Phi(\xi, a)$ at a degree at most N depends only on P_N , so $D_N \geq 0$. It is precisely the finite prefix of the true recurrence defect $\Phi(\xi, a) - a$.

The signed equation

$$B_N = \Phi(\xi, B_N) - D_N \quad (10)$$

is a convenient shorthand, but its right-hand side is not a positive polynomial in ξ . The correct positivity argument uses $Y_N = B_N - P_N$ and the map

$$\begin{aligned}\Psi_N(\xi, Y) &= \Phi(\xi, P_N + Y) - T_N \\ &= \Phi(\xi, P_N + Y) - \Phi(\xi, P_N) + [\xi^{>N}] \Phi(\xi, P_N).\end{aligned}\tag{11}$$

Every coefficient of Ψ_N in $\xi, Y_1, \dots, Y_d$ is nonnegative. Indeed, in the difference on the second line, the terms independent of Y cancel, leaving only monomials containing at least one Y . Moreover, $\Psi_N(0, 0) = 0$ and $(\Psi_N)_Y(0, 0) = L$.

Theorem 3.3 (Prefix-corrected majorant). *The map Ψ_N has a unique fixed point $Y_N \in \xi \mathbb{R}_{\geq 0}[[\xi]]^d$. Its coefficients through degree N vanish. The series $B_N = P_N + Y_N$ satisfies*

$$a \leq B_N, \quad [\xi^{\leq N}] B_N = P_N,$$

and solves (10).

Proof. The fixed-point statement follows from lemma 3.2. The constant-in- Y part of Ψ_N begins above degree N , so degree induction gives $[\xi^n] Y_N = 0$ for $n \leq N$. Put $U = a - P_N \geq 0$. The low-degree part of $\Phi(\xi, a) - a$ is D_N , and its remaining coefficients are nonnegative. Therefore

$$U \leq \Phi(\xi, P_N + U) - T_N = \Psi_N(\xi, U).$$

Formal comparison yields $U \leq Y_N$. Adding P_N proves domination. Substituting the fixed-point equation for Y_N yields (10). $\square$

Theorem 3.4 (Rational upper certificate). *Let $\zeta > 0$. Suppose a vector $v \in \mathbb{R}^d$ satisfies*

$$v \geq P_N(\zeta), \quad v \geq \Phi(\zeta, v) - D_N(\zeta).\tag{12}$$

Then all coordinates of B_N converge at ζ , and

$$B_N(\zeta) \leq v, \quad a_{i,n} \leq v_i \zeta^{-n}.$$

For the polyomino system this implies $A_n \leq v_G \zeta^{-n}$ and $\lambda \leq \zeta^{-1}$. When the input prefix, ζ , and v are rational, verification of (12) requires rational arithmetic only.

Proof. Let $w = v - P_N(\zeta) \geq 0$. Equations (9) and (11) show that (12) is equivalent to

$$\Psi_N(\zeta, w) \leq w.$$

The numerical iterates $w^{(0)} = 0$, $w^{(j+1)} = \Psi_N(\zeta, w^{(j)})$ therefore stay between 0 and w . The corresponding formal iterates are polynomials with nonnegative coefficients. Their evaluation at ζ is exactly $w^{(j)}$. For every fixed degree M , a sufficiently late formal iterate agrees with Y_N through degree M by lemma 3.2. Hence

$$\sum_{n=0}^M [\xi^n] Y_N \zeta^n \leq w.$$

Monotone convergence of these bounded finite sums proves $Y_N(\zeta) \leq w$. Adding $P_N(\zeta)$ proves the asserted bound for B_N . Every nonnegative term of a bounded series is bounded by its sum, yielding the coefficient estimate. Now use (5) and (3). $\square$

Remark 3.5 (The tail condition is essential). A solution of $v \geq \Phi(\zeta, v) - D_N(\zeta)$ alone need not be a valid certificate. The subtraction can create extraneous branches or vectors below the known prefix. The separate inequality $v \geq P_N(\zeta)$ is part of the proof and is checked explicitly by the verifier.

3.3 Monotonicity of the hierarchy

Theorem 3.6 (More exact data never worsens the majorant). *For every $N \geq 1$,*

$$a \leq B_{N+1} \leq B_N.$$

Thus the radii of convergence of the corrected majorants are nondecreasing, and their reciprocal exponential-growth bounds are nonincreasing. Any real upper certificate for prefix N is also an upper certificate for prefix $N + 1$ at the same evaluation point.

Proof. The first inequality is [theorem 3.3](#). Through degree N , the two majorants agree. At degree $N + 1$, B_{N+1} has the true coefficient and B_N has a coefficient at least as large. Above that degree, both obey the same positive coefficient recurrence, because the defect polynomials have no terms there. Degree induction using $(I - L)^{-1} \geq 0$ proves the second inequality. Coefficientwise domination implies the radius assertion.

If v is an N -certificate, then the preceding theorem and $a \leq B_N$ show that $P_{N+1}(\zeta) \leq v$. Also $D_{N+1} - D_N$ has nonnegative coefficients, so $\Phi(\zeta, v) - D_{N+1}(\zeta) \leq v$. Both certificate conditions for $N + 1$ follow. $\square$

For the present system the defect is zero through size 6. Its size-7 coefficient, in the order (4), is

$$[\xi^7]D_7 = (0, 1, 2, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 2). \quad (13)$$

For example, $a_{D,7} = 260$, whereas the deletion bound uses $a_{G,6} = 261$; the corresponding defect coefficient is 1. This first discrepancy explains why prefix lengths 1 through 6 do not improve the original equality system.

4 An exact upper bound below 4.5

4.1 The certificate

Take $N = 18$ and

$$\zeta = \frac{500}{2249}.$$

Use the exact marked counts in [Appendix A](#) to construct P_{18}, T_{18}, D_{18} by (9). The vector v is given in [table 2](#); every numerator in its second column has common denominator 10^9 .

Define the residual and tail budget by

$$\sigma_i = v_i - \Phi_i(\zeta, v) + D_{18,i}(\zeta), \quad \eta_i = v_i - P_{18,i}(\zeta). \quad (14)$$

The last two columns are conservative lower bounds:

$$\sigma_i \geq \frac{\text{third-column entry}}{10^{12}}, \quad \eta_i \geq \frac{\text{fourth-column entry}}{10^6}.$$

They are obtained by flooring the exact rational numbers after scaling; no printed decimal approximation is being used to decide a sign.

Theorem 4.1 (Computer-assisted polyomino bound). *For the fixed-polyomino counting convention in [section 2](#),*

$$A_n \leq \frac{927884613}{10^9} \left(\frac{2249}{500} \right)^n \quad (n \geq 1), \quad \lambda \leq 4.498 < 4.5.$$

Coordinate	Numerator of v_i	10^{12} -scaled slack	10^6 -scaled tail
c	347,752,792	99,824	28,076
d	427,615,666	100,303	59,180
e	564,832,205	100,345	123,441
f	1,551,872,652	100,000	542,918
g	927,884,613	99,000	260,176
h	727,305,121	100,000	159,091
p	623,987,939	99,749	282,741
q	363,052,309	100,459	136,735
r	234,092,913	100,000	63,133
s	299,689,355	99,657	99,910
t	283,350,363	100,000	89,179
u	487,086,219	100,056	192,950
v	119,025,613	99,875	52,209
w	97,883,000	99,767	38,611
x	164,324,650	100,195	36,970
y	136,209,813	99,861	24,521
z	112,439,657	100,134	14,724

Table 2: The rational size-18 upper certificate and exact lower bounds on its margins. All vector entries have denominator 10^9 .

Proof. The finite enumeration described below produces the true coefficients of P_{18} in Appendix A. Bui’s recurrence lemma gives (7). Exact integer polynomial convolution constructs D_{18} . Substituting the rational vector of table 2 into (14) gives

$$\min_i \sigma_i \geq \frac{99}{10^9} > 0, \quad \min_i \eta_i \geq \frac{14724}{10^6} > 0.$$

Thus theorem 3.4 applies. Its G coordinate is $v_G = 927884613/10^9$, yielding the claimed inequalities. $\square$

The strict residuals also show that the certificate is not balanced on a rounding boundary. By continuity, the same vector works for a sufficiently small increase of ζ . We retain 2249/500 as the stated bound because it is simple and explicitly verified.

4.2 Discovery is not verification

The search for a candidate used floating-point solutions of

$$\Phi(\zeta, v) - D_N(\zeta) = v, \quad \det(I - \Phi_X(\zeta, v)) = 0.$$

These equations suggest where the equality envelope becomes critical. For a slightly smaller evaluation point, we solved a perturbed fixed-point equation with a small positive residual in every coordinate, then rounded the candidate to denominator 10^9 . Such a solver may converge to an undesired branch or misestimate a threshold. Neither event can invalidate an accepted certificate: the verifier checks the two inequalities in (12) independently from scratch.

Table 3 records the observed progression. The middle column contains numerical critical-point estimates, not proved optimal values or certified lower bounds. Every entry in the last column has a separate exact rational certificate in the archive.

The original recurrence’s growth is already within 10^{-5} of the repository’s bound, as section 7 will prove. Consequently, theorem 4.1 does not arise from finding a better supersolution for the unchanged system. The correction changes the equality envelope while preserving its validity as an upper bound on the true counts.

Prefix size N	Estimated equality growth	Certified upper bound
1–6	4.523499228	4.5235 (original certificate)
7	4.523230947	4.5233
8	4.522434056	4.5225
9	4.521121227	4.5212
10	4.519368978	4.5194
11	4.517262575	4.5173
12	4.514881253	4.5149
13	4.512294495	4.5123
14	4.509560290	4.5096
15	4.506725656	4.5068
16	4.503828169	4.5039
17	4.500897596	4.5009
18	4.497957351	4.4980

Table 3: Progress under exact-prefix correction. Numerical estimates and rigorous upper certificates are deliberately separated.

4.3 A reproducible finite enumeration

Every translation class has a unique representative whose leftmost cell on the bottom row is $(0, 0)$. All its cells then lie in

$$\mathcal{H} = \{(x, y) : y > 0\} \cup \{(x, 0) : x \geq 0\}.$$

The main enumerator maintains a connected occupied set S , a finite ordered frontier U of available adjacent cells, and an excluded set E . Initially $S = \emptyset$, $U = \{(0, 0)\}$, and $E = \emptyset$. At a call, take the last frontier cell p , include it in S , record the resulting object, and recurse after adding newly available neighbors. After that branch returns, remove p and exclude it for all later branches of the same call. The exclusions introduced by a call are removed when that call returns. No quotient by rotations or reflections is performed.

Proposition 4.2 (Enumeration correctness). *The include/exclude frontier procedure records each nonempty finite edge-connected subset of $\mathcal{H}$ containing $(0, 0)$ exactly once, up to the specified size limit. Thus its totals are precisely the fixed-polyomino counts.*

Proof. Every recorded set is connected because each added cell is adjacent to the previous occupied set, except for the initial origin. It stays in $\mathcal{H}$ by the admissibility test.

Fix a target set P of the required kind. At any intermediate state with $S \subsetneq P$ and no excluded cell in P , connectedness implies that some cell of $P \setminus S$ lies in the available frontier. In the order processed by the current call, there is a unique first frontier cell belonging to P . All cells processed before it can be excluded without affecting P . The branch including that unique cell preserves the same invariant for P . Induction reaches $S = P$, which is recorded.

The same argument proves uniqueness: the first processed frontier cell belonging to P determines the only branch that can contain P without excluding one of its cells. Applying this observation recursively determines a unique path. Finally, the chosen anchor gives exactly one representative of each translation class. $\square$

For an occupied anchor, its membership in each pattern depends only on ten nearby positions:

$$(-1, 1), (1, 1), (2, 1), (-1, 0), (1, 0), (2, 0), (-1, -1), (0, -1), (1, -1), (2, -1).$$

These are encoded by a ten-bit mask. All 2^{10} masks are preclassified against the required and forbidden sets of [table 1](#). The direct implementation scans all occupied anchors whenever an

object is recorded. The faster implementation maintains the pattern totals under cell insertion and deletion. Inserting p changes only the contribution of p and those anchors $p - o$ for which o is one of the ten offsets. Removing p reverses exactly these changes. This local invariant proves that incremental updates give the same occurrence count as a complete scan.

The size-18 computation records

$$A_{18} = 1,540,820,542, \quad \sum_{n=1}^{18} A_n = 2,083,404,030$$

objects across all recorded sizes. The unmarked totals through 18 agree with OEIS A001168 [12]; those familiar totals are a consistency check, not a replacement for the marked enumeration.

4.4 Finite-machine safety and independent checks

The C++ sources are restricted to size at most 18. Any cell of an anchored n -cell polyomino satisfies $|x| \leq n - 1$ and $0 \leq y \leq n - 1$. The padded array is wide and tall enough for these coordinates and for all mask and neighbor offsets. Thus no boundary wrapping or out-of-range access is needed by the algorithm.

The counters are unsigned 64-bit integers. An a priori bound verifies that overflow is impossible at these sizes. A deterministic depth-first spanning tree of an anchored n -cell set can be encoded by a plane rooted tree and directions for its $n - 1$ forward steps. There are Cat_{n-1} plane tree shapes. The first forward step has at most four choices; every later forward step has at most three, since a nonroot vertex cannot move back to its parent and the root has already used one child direction. Consequently, for $n \geq 2$,

$$A_n \leq 4 \cdot 3^{n-2} \text{Cat}_{n-1}.$$

This overcounts heavily but is sufficient. Every marked count is at most nA_n , and the largest bound for $n \leq 18$ is

$$18 \cdot 4 \cdot 3^{16} \text{Cat}_{17} = 401816383504818480 < 2^{64}.$$

Incremental counters count occurrences in the currently occupied finite set; the insertion/deletion invariant also prevents underflow.

Two structurally different enumeration approaches were compared through size 11: the C++ include/exclude procedure and a Python procedure that repeatedly adds each boundary cell, translates each shape to a canonical coordinate tuple, and deduplicates using a set. All 187 marked values and all unmarked totals agreed. Direct scanning and incremental scanning in C++ agreed through size 16. The final size-18 output also agrees with the previous size-16 table on every earlier entry.

These checks do not amount to independent formal certification of all size-18 values. They locate the remaining computational trust boundary: the correctness of the supplied enumerator, its implementation, and its execution. The recurrence inequalities themselves are not learned from this table. Five exact partition identities and all nonnegative recurrence defects are checked again by the rational verifier.

[write] *Regeneration at intake*. The delivered validation record says that no fresh size-18 run was made after the enumerators' final command-line changes. When the manuscript was received by ProveIt, the delivered code/enumerate.cpp was compiled afresh (c++ -O3 -std=c++17, g++ 16.1.0) and run as `enumerate 18` (8 min 23 s on one core of the intake machine); after CSV parsing its output equals the shipped data/profiles_18.csv in all 19 columns of all 18 rows (the files differ only in line endings). Fresh runs through sizes 11–14, 16 and 17 matched as well, and an independent set-based Python enumerator, with pattern coordinates taken from the project's `Patterns.lean` rather than from this package, matched all 10 unmarked and 170 marked entries

through size 10. So the size-18 table is now reproduced from the final source. This does not change the boundary stated above: the regeneration uses the same program and algorithm, the rational verifier checks arithmetic and necessary consistency conditions only, and the correctness of the enumeration rests on [proposition 4.2](#) and the incremental-update argument together with this regeneration.

5 Balanced-monomial certificates of divergence

5.1 The certificate format

Consider a finite positive polynomial system

$$\Phi_i(\zeta, X) = \sum_{k \in K_i} c_{ik} \zeta^{b_{ik}} X^{\alpha_{ik}}, \quad c_{ik} \in \mathbb{Q}_{>0}, \quad b_{ik} \in \mathbb{N}, \quad \alpha_{ik} \in \mathbb{N}^d. \quad (15)$$

Fix a rational $\zeta > 0$. We wish to certify that there is no $X \in \mathbb{R}_{>0}^d$ satisfying $\Phi(\zeta, X) \leq X$.

Assign a nonnegative integer weight m_{ik} to each monomial, not all zero, and define row sums

$$r_i = \sum_{k \in K_i} m_{ik}.$$

The weights are *balanced* if

$$\sum_{i,k} m_{ik} \alpha_{ik,j} = r_j \quad (1 \leq j \leq d). \quad (16)$$

Define the positive rational product

$$C(\zeta, m) = \prod_{i,k:m_{ik}>0} \left(\frac{c_{ik} \zeta^{b_{ik}} r_i}{m_{ik}} \right)^{m_{ik}}. \quad (17)$$

A row of weight zero contributes no factor. The entire certificate consists of ζ , the integer weights, and a verification that $C(\zeta, m) > 1$.

Theorem 5.1 (Balanced-monomial obstruction). *If (16) holds and $C(\zeta, m) > 1$, then no positive supersolution of $\Phi(\zeta, X) \leq X$ exists.*

Proof. Suppose such a supersolution exists. For each row with $r_i > 0$, put $p_{ik} = m_{ik}/r_i$ on the positive-weight terms. Weighted arithmetic–geometric mean gives

$$\begin{aligned} 1 &\geq \frac{\Phi_i(\zeta, X)}{X_i} \geq \sum_{k:m_{ik}>0} c_{ik} \zeta^{b_{ik}} X^{\alpha_{ik}-e_i} \\ &\geq \prod_{k:m_{ik}>0} \left(\frac{c_{ik} \zeta^{b_{ik}} X^{\alpha_{ik}-e_i}}{p_{ik}} \right)^{p_{ik}}. \end{aligned}$$

Raise the inequality to r_i and multiply over rows. The exponent of X_j in the resulting right-hand side is $\sum_{i,k} m_{ik} \alpha_{ik,j} - r_j = 0$ by balance. All unknown variables cancel, leaving $1 \geq C(\zeta, m) > 1$, a contradiction. $\square$

The cancellation is the essential point. A certificate does not need to supply a very large iterate, an approximate divergent orbit, or an approximate critical point. Its inequalities quantify over every positive candidate supersolution at once.

Example 5.2 (Catalan recurrence). For $\Phi(\xi, x) = \xi + x^2$, assign weight one to both terms. Then the incoming exponent is $0 + 2 = 2 = r$, and

$$C(\zeta, m) = (2\zeta)(2) = 4\zeta.$$

Thus $\zeta > 1/4$ immediately has a two-weight divergence certificate, recovering the usual Catalan threshold from one arithmetic–geometric mean inequality.

5.2 From nonexistence to a lower growth bound

Let B be the least formal fixed point of a proper positive system. If its coordinates converge at ζ , finite products of nonnegative convergent series can be evaluated, giving

$$B(\zeta) = \Phi(\zeta, B(\zeta)).$$

If the system is productive in the sense defined in the next section, this fixed point is strictly positive. Therefore [theorem 5.1](#) implies that at least one coordinate fails to converge at ζ .

Define the system’s exponential growth by

$$\mu = \max_i \limsup_{n \rightarrow \infty} ([\xi^n] B_i)^{1/n}.$$

Equivalently, $1/\mu$ is the minimum of the coordinate radii of convergence. Cauchy–Hadamard then gives

$$\mu \geq \zeta^{-1}. \quad (18)$$

We use the limsup formulation so that no unproved assertion about periodicity or existence of individual root limits is needed.

For productive systems with a strongly connected dependency graph, the coordinate radii coincide. Indeed, for every dependency edge $i \rightarrow j$, choose one positive coefficient in each of the other factors of a monomial containing X_j . The formal fixed-point equation implies $B_i \geq c \xi^k B_j$ for some $c > 0$ and integer $k \geq 0$. Thus $R_i \leq R_j$. Following paths in both directions proves equality.

Do not reverse the majorant inequality. The true polyomino series is below the equality solution, so an upper bound on equality growth bounds λ from above. A lower bound on equality growth does not bound λ from below. The dual certificate is useful for certifying the limitation of a chosen recurrence model and for answering the recurrence-growth question, not for reversing this implication.

6 Completeness of the certificate alternative

6.1 The hypotheses

For this section fix $\zeta > 0$ and write $F(X) = \Phi(\zeta, X)$. The coefficients of F are positive on its listed monomials.

Definition 6.1. The map F is *productive* if some finite iterate of F , starting from zero, has every coordinate strictly positive. Its dependency graph has an edge $i \rightarrow j$ if some monomial of F_i contains X_j . The map is *strongly connected* if this graph is strongly connected, and *nonlinear* if at least one monomial has total degree at least two in the unknowns.

Productivity is stronger than merely having a constant term somewhere. For example, $F_1 = 1 + X_1 X_2$, $F_2 = X_1 X_2$ has a strongly connected graph, but iteration from zero never makes X_2 positive. This degeneracy is excluded. All three conditions are finite structural checks on the monomial list; they hold for (6) and are replayed by the verifier.

6.2 Compactness of bounded multiplicative residuals

Lemma 6.2. *If F is productive, strongly connected, and nonlinear, then, for every $K > 0$, the set*

$$S_K = \{X \in \mathbb{R}_{>0}^d : F(X) \leq KX\}$$

is contained in a compact box $[a, b] \subset \mathbb{R}_{>0}^d$ and is closed in that box. In particular, it is compact, possibly empty.

Proof. Every $X \in S_K$ satisfies $X \geq K^{-1}F(X)$. Iteration from zero of the map $K^{-1}F$ therefore stays below X . Multiplication by a positive scalar does not change which coordinates become positive at any iteration, so productivity supplies a fixed vector $a > 0$ with $X \geq a$ for every $X \in S_K$.

For each edge $i \rightarrow j$, choose a monomial containing X_j . Lower-bound all its other factors, including extra copies of X_j if necessary, by the corresponding coordinates of a . This gives a constant $c_{ij} > 0$ such that

$$F_i(X) \geq c_{ij}X_j \quad (X \geq a).$$

On S_K it follows that $X_i \geq (c_{ij}/K)X_j$. Strong connectivity now gives constants $u_i, v_i > 0$ with

$$u_iX_1 \leq X_i \leq v_iX_1 \quad (X \in S_K).$$

Choose a nonlinear monomial of total degree $q \geq 2$ in row h . The lower comparisons give $F_h(X) \geq cX_1^q$ for some $c > 0$, whereas $F_h(X) \leq KX_h \leq Kv_hX_1$ on S_K . Thus $X_1^{q-1} \leq Kv_h/c$, a uniform upper bound. The other coordinates are then bounded above by their comparisons with X_1 . Finally, the polynomial inequalities are closed within this positive compact box. $\square$

6.3 An elementary convex proof

Put $X_j = e^{y_j}$ and define

$$f_i(y) = \log F_i(e^y) - y_i, \quad h(y) = \max_i f_i(y). \quad (19)$$

Each f_i is a log-sum-exp of affine functions and is convex. For completeness, its Hessian is the covariance matrix of the monomial exponent vectors under their positive softmax probabilities, and is therefore positive semidefinite. Sublevel sets of h correspond to the sets S_K of [lemma 6.2](#). Hence h attains its global minimum.

Theorem 6.3 (Complete rational certificate alternative). *Suppose F has rational coefficients and is productive, strongly connected, and nonlinear. Exactly one of the following holds:*

- (i) *there is $X > 0$ with $F(X) \leq X$;*
- (ii) *there are balanced nonnegative integer monomial weights, not all zero, for which the product (17) is greater than one.*

Here the coefficient $c_{ik}\zeta^{b_{ik}}$ in (17) is simply the corresponding coefficient of F .

Proof. Mutual exclusion is [theorem 5.1](#). Suppose (i) fails. Let y^* minimize h , and put $\delta = h(y^*)$. Since a minimum is attained and no point has $h \leq 0$, we have $\delta > 0$.

Let I be the active rows, those with $f_i(y^*) = \delta$. The origin belongs to the convex hull of their gradients. To see this without invoking an optimization duality theorem, suppose otherwise and let g be the point of that compact convex hull nearest the origin. Then $g \neq 0$ and $g \cdot \nabla f_i(y^*) \geq \|g\|^2$ for every active row. A sufficiently small step in direction $-g$ strictly decreases every active f_i .

The inactive rows remain below the old maximum by continuity and finiteness. This contradicts minimality of h . Consequently there are numbers $\theta_i \geq 0$, supported on I , such that

$$\sum_i \theta_i = 1, \quad \sum_i \theta_i \nabla f_i(y^*) = 0.$$

For each monomial let

$$p_{ik} = \frac{c_{ik} \zeta^{b_{ik}} e^{\alpha_{ik} \cdot y^*}}{F_i(e^{y^*})}, \quad \beta_{ik} = \theta_i p_{ik}.$$

Then $\sum_k \beta_{ik} = \theta_i$, and the gradient identity is exactly

$$\sum_{i,k} \beta_{ik} \alpha_{ik,j} = \theta_j.$$

Thus the real weights β are balanced and have total mass one. Their logarithmic product is

$$\begin{aligned} \mathcal{L}(\beta) &= \sum_{i,k:\beta_{ik}>0} \beta_{ik} \log \left(\frac{c_{ik} \zeta^{b_{ik}} \theta_i}{\beta_{ik}} \right) \\ &= \sum_i \theta_i \log F_i(e^{y^*}) - \sum_j \theta_j y_j^* \\ &= \sum_i \theta_i f_i(y^*) = \delta > 0. \end{aligned}$$

It remains to replace real weights by integers. Restrict to the support of β . On this support, balance and total mass one form a consistent linear system with rational coefficients. Gaussian elimination supplies a rational particular solution and a rational basis for the homogeneous solution space. Approximating the real free parameters by rationals gives balanced rational weights arbitrarily close to β , positive on the same support. The displayed logarithmic expression is continuous there, so a sufficiently close rational approximation still has positive value. Multiplying by a common denominator gives integer weights. Scaling multiplies the logarithmic product by that positive denominator, preserving strict positivity. This proves (ii). $\square$

6.4 What this settles, and what it does not

A positive numerical supersolution bounds the zero iterates and hence all formal partial sums, by the same argument as in [theorem 3.4](#). Thus it forces convergence at the evaluation point. For a proper productive system in the stated class, divergence at a rational point therefore has a finite balanced-monomial obstruction, and a finite obstruction implies the lower-growth bound (18). Thus there is a certificate alternative, not just a test based on watching a floating-point iteration grow. This answers the certificate-existence aspect of Bui's Question 2 for this class, including his seventeen-variable system.

The result does not assert that the original question for every conceivable convolution, maximum, or signed recurrence has been settled. Nor does it give a uniform polynomial bound on the bit length of the weights as a target approaches the exact growth threshold. The explicit certificate below is small and quickly checked; a worst-case certificate-size theory remains a separate problem. The underlying convex mechanism is classical geometric programming, here specialized and proved in an arithmetic certificate form [13, 14, 15].

7 A compact dual certificate for the original recurrence

7.1 The integer weights

For the uncorrected map (6), take

$$\zeta_{\text{dual}} = \frac{100000}{452349}.$$

Use the weights in table 4. There are 53 positive integers, with sum 1,000,000 and largest entry 150,463. Direct integer addition verifies all seventeen balance equations (16).

Row	r_i	Monomial weights, in the order in (6)
c	4,857	3,080, 1,777
d	57,691	29,588, 28,103
e	242,698	93,305, 149,393
f	149,393	88,617, 60,776
g	248,519	150,463, 98,056
h	47,024	27,413, 19,611
p	60,776	39,670, 15,071, 3,712, 1,589, 734
q	113,127	63,384, 36,448, 7,450, 4,066, 1,779
r	6,370	3,666, 2,704
s	27,124	18,469, 6,430, 1,247, 679, 299
t	6,546	3,753, 2,793
u	11,651	7,354, 3,067, 759, 322, 149
v	4,382	2,413, 1,560, 289, 120
w	3,026	2,033, 796, 139, 58
x	8,357	4,786, 2,331, 1,240
y	7,427	4,155, 2,506, 766
z	1,032	702, 256, 74

Table 4: An explicit balanced-monomial certificate. The coefficient of every monomial is one. Within each coordinate row, weights follow the term order in (6).

For example, the incoming C exponent is contributed only by the ξc terms in rows Y and Z . Their weights are 4155 and 702, whose sum is $4857 = r_C$. Other balance equations count repeated variables with multiplicity: the z^2 in row U contributes twice its weight to the incoming Z exponent.

Exact rational interval evaluation proves

$$\frac{1000668023141660979}{10^{18}} \leq \log C(\zeta_{\text{dual}}, m) \leq \frac{1000668023155325499}{10^{18}}. \quad (20)$$

In particular, $1 < \log C < 1001/1000$, so $C > 1$. The slightly generous unit-sized lower margin is merely a convenient verification statement; scaling all weights scales the logarithm.

Theorem 7.1 (Certified limitation of the original system). *The growth μ_{original} of the original equality recurrence satisfies*

$$\frac{452349}{100000} \leq \mu_{\text{original}} \leq \frac{9047}{2000}.$$

Consequently, no supersolution of the original system can establish a growth upper bound smaller than 4.52349.

Proof. Balance and (20) give the lower bound by theorem 5.1 and (18). The repository’s rational supersolution at $\zeta = 2000/9047$, replayed by our verifier without a defect correction, gives the upper bound. Finally, a supersolution at a smaller claimed growth value would imply a radius larger than $100000/452349$, contradicting the certified lower bound. $\square$

The last statement can also be read directly: if a supersolution existed at some $\zeta' > \zeta_{\text{dual}}$, monotonicity of Φ in its first argument would make it a supersolution at ζ_{dual} , which the dual certificate forbids. This is an exact obstruction, not a conclusion drawn from a numerical optimizer failing to improve the bound.

7.2 Exact verification without enormous integer products

Computing the rational product (17) literally is possible in principle, but raising many rational factors to large integer weights is unnecessary. Instead, enclose its logarithm using rational arithmetic. For $1 \leq q \leq 2$, let $u = (q - 1)/(q + 1)$, so $0 \leq u \leq 1/3$. The identity

$$\log q = 2 \sum_{k \geq 0} \frac{u^{2k+1}}{2k+1}$$

follows by integrating the geometric series for $2/(1 - u^2)$ from 0 to u . The remainder after M terms satisfies

$$0 \leq \log q - 2 \sum_{k=0}^{M-1} \frac{u^{2k+1}}{2k+1} \leq \frac{2u^{2M+1}}{(2M+1)(1-u^2)}. \quad (21)$$

All endpoints are rational when q is rational.

For a general positive rational q , write $q = 2^e w$ with integer e and $1 \leq w < 2$. Evaluate $\log w$ and $\log 2$ by (21); multiply the interval for $\log 2$ by e , reversing endpoint roles when $e < 0$. This gives a rational enclosure for $\log q$.

The supplied verifier uses $M = 16$. Each individual logarithm interval is rounded outward to denominator 10^{18} before multiplication by its integer weight. This optional outward rounding prevents huge denominators when intervals from many factors are added. It cannot invalidate enclosure. Adding the resulting weighted intervals gives exactly (20). The same cached enclosure of $\log 2$ can be reused; there are at most 54×16 short series terms to evaluate.

The verification cost depends on the bit sizes of the rational monomial factors, weights, and chosen precision, rather than on the number of iterates required to witness numerical divergence. No logarithm supplied by a floating-point library enters the proof. We make no universal claim that certificate discovery or certificate size is polynomial in the reciprocal distance to the growth threshold.

7.3 How the weights were discovered

The following construction is useful when a finite positive critical point has been found numerically. Suppose

$$\tau = \Phi(\rho, \tau), \quad J = \Phi_X(\rho, \tau), \quad \ell^T J = \ell^T, \quad \ell > 0.$$

Normalize by $C = \sum_i \ell_i \tau_i$ and define

$$\beta_{ik} = \frac{\ell_i c_{ik} \rho^{b_{ik}} \tau^{\alpha_{ik}}}{C}.$$

The row sums are $\ell_i \tau_i / C$, and the left-eigenvector identity shows that the weights are balanced. At $\zeta = \rho$, their logarithmic product is zero. At $\zeta > \rho$, it becomes

$$\left(\sum_{i,k} \beta_{ik} b_{ik} \right) \log(\zeta/\rho) > 0$$

provided at least one positively weighted term contains the size variable. Rational approximation inside the balance equations then yields integer weights. This is how the displayed certificate was found.

The approximate critical point and eigenvector are not inputs to the final verifier. Only the resulting integers and rational evaluation point are. The completeness proof in [section 6](#) also does not require that a numerical critical point be correct or even available.

8 The limits of the exact-prefix hierarchy

8.1 A spectral criterion

Fix a positive evaluation point ζ at which the true series $\mathbf{a}$ converges coordinatewise. Put

$$\mathbf{a} = \mathbf{a}(\zeta), \quad p_N = P_N(\zeta), \quad J = \Phi_X(\zeta, \mathbf{a}), \quad J_N = \Phi_X(\zeta, p_N),$$

and define the nonnegative forcing vector

$$b_N = ([\xi^{>N}] \Phi(\xi, P_N(\xi)))(\zeta).$$

At this evaluation point,

$$\Psi_N(\zeta, y) = b_N + \Phi(\zeta, p_N + y) - \Phi(\zeta, p_N). \quad (22)$$

We have $p_N \uparrow \mathbf{a}$ and $J_N \rightarrow J$. Moreover, $b_N \rightarrow 0$: coefficientwise $\Phi(\xi, P_N) \leq \Phi(\xi, \mathbf{a})$, and the latter series converges at ζ . Its high-degree tail bounds b_N .

Theorem 8.1 (Strict stability and obstruction). *Assume the preceding setup.*

- (a) *If $\text{spr}(J) < 1$, then all sufficiently large N admit a real upper certificate at ζ . If the polynomial data and ζ are rational, the certificate can be chosen rational.*
- (b) *For any N , if J_N is irreducible, $\text{spr}(J_N) \geq 1$, and $b_N \neq 0$, then no upper certificate at ζ exists for that N .*
- (c) *If $\text{spr}(J) > 1$, J_N is irreducible for all sufficiently large N , and $b_N \neq 0$ for all sufficiently large N , then no finite-prefix member of the hierarchy admits an upper certificate at ζ .*

Proof. For (a), the nonnegative matrix geometric series gives

$$h = (I - J)^{-1} \mathbf{1} > 0, \quad Jh = h - \mathbf{1}.$$

For sufficiently small $\varepsilon > 0$, polynomial expansion in the nonnegative direction εh gives, uniformly for $0 \leq p_N \leq \mathbf{a}$,

$$\Phi(\zeta, p_N + \varepsilon h) - \Phi(\zeta, p_N) \leq \varepsilon Jh + C\varepsilon^2 \mathbf{1}$$

for some finite $C \geq 0$. Choose ε with $C\varepsilon^2 \leq \varepsilon/4$, and then N sufficiently large that $b_N \leq (\varepsilon/4)\mathbf{1}$. Equation (22) yields

$$\Psi_N(\zeta, \varepsilon h) \leq \varepsilon h - (\varepsilon/2)\mathbf{1}.$$

Thus $p_N + \varepsilon h$ is a certificate with strict slack and strictly positive tail budget. When the data and evaluation point are rational, a sufficiently close rational vector preserves both strict inequalities.

For (b), nonnegative polynomial expansion gives $\Psi_N(\zeta, y) \geq b_N + J_N y$ for $y \geq 0$. Let $\ell > 0$ be a left Perron vector of the irreducible matrix J_N , with eigenvalue $r = \text{spr}(J_N) \geq 1$. A certificate would give $y \geq b_N + J_N y$, and hence

$$(1 - r)\ell^T y \geq \ell^T b_N > 0,$$

which is impossible.

For (c), continuity of spectral radius gives $\text{spr}(J_N) > 1$ for all sufficiently large N . Part (b) rules out certificates at all those levels. By [theorem 3.6](#), an earlier certificate would remain a certificate at all later levels. Therefore no earlier one exists either. $\square$

The equality case $\text{spr}(J) = 1$ is deliberately not classified in general. The nonzero forcing and irreducibility conditions in the obstruction are not decorative: without them a critical linear map can have supersolutions. For the Bui map, positive prefix coordinates make the dependency pattern irreducible, and the C row already supplies a nonzero high-degree forcing term whenever the top E coefficient is positive.

8.2 An exact counterexample to asymptotic exactness

Theorem 8.2 (All finite prefixes may retain a growth gap). *There is a scalar proper positive convolution inequality with a true series of growth 1 for which the exact-prefix majorants have growth tending to 3, not 1.*

Proof. Take

$$A(\xi) = \frac{\xi}{1-\xi} = \sum_{n \geq 1} \xi^n, \quad \Phi(\xi, x) = \xi + x^2.$$

The coefficient inequality $A \leq \xi + A^2$ holds: the coefficient of ξ^n in A^2 is $n-1$ for $n \geq 2$. The exact-prefix defect is

$$D_N(\xi) = \sum_{n=3}^N (n-2)\xi^n,$$

with an empty sum for $N < 3$. The corrected equality equation is

$$B_N = \xi + B_N^2 - D_N(\xi). \quad (23)$$

Let $c_N(t) = t - D_N(t)$ for real t . Since

$$D_\infty(t) = \frac{t^3}{(1-t)^2},$$

we have $D_\infty(1/3) = 1/12$ and $D'_\infty(t) \leq 1$ on $0 \leq t \leq 1/3$, with equality only at the right endpoint. A finite truncation has strictly smaller derivative there. Thus c_N is strictly increasing on this interval, $c_N(1/4) \leq 1/4$, and $c_N(1/3) > 1/4$. There is a unique

$$\rho_N \in [1/4, 1/3) \quad \text{with} \quad c_N(\rho_N) = 1/4.$$

At $t = \rho_N$, the value $v = 1/2$ solves (23), and

$$P_N(\rho_N) < A(\rho_N) < 1/2.$$

It is therefore a valid upper certificate. For any $t \in (\rho_N, 1/3]$, the discriminant of $x^2 - x + c_N(t) = 0$ is negative, so there is no finite real fixed point. Hence the radius of B_N is exactly ρ_N .

The truncated polynomials D_N converge uniformly to D_∞ on $[0, 1/3]$. Every limit point ρ of the sequence ρ_N therefore satisfies

$$\rho - \frac{\rho^3}{(1-\rho)^2} = \frac{1}{4}, \quad \text{equivalently} \quad (3\rho - 1)^2 = 0.$$

Thus $\rho_N \rightarrow 1/3$, and the corrected growth tends to 3. The true series has radius 1 and growth 1. $\square$

The same barrier is visible directly in [theorem 8.1](#):

$$\Phi_x(t, A(t)) = \frac{2t}{1-t}$$

crosses one at $t = 1/3$, well before the true singularity at $t = 1$. Every finite coefficient of B_N eventually agrees with A , but radii of convergence do not pass continuously through this coefficientwise limit.

This counterexample is not a disproof of Bui's separate Question 1 about increasing neighborhood size with improved geometric recurrences. Here the local polynomial map is held fixed and only a finite exact prefix is substituted. Those are different hierarchies. It does show that an eventual convergence claim for the present prefix method requires a stability argument or additional geometric refinement.

8.3 A conditional sensitivity formula for choosing improvements

A useful guide for future searches is available when a smooth critical branch has already been justified. Suppose an admissible defect improvement $D + \varepsilon E$ has critical data $(\rho(\varepsilon), \tau(\varepsilon))$ satisfying

$$\Phi(\rho(\varepsilon), \tau(\varepsilon)) - D(\rho(\varepsilon)) - \varepsilon E(\rho(\varepsilon)) = \tau(\varepsilon).$$

At $\varepsilon = 0$, let $\ell^T J = \ell^T$ for $J = \Phi_x(\rho, \tau)$. Differentiating and multiplying by ℓ^T cancels the unknown derivative of τ , yielding

$$\rho'(0) = \frac{\ell^T E(\rho)}{\ell^T (\Phi_\xi(\rho, \tau) - D'(\rho))}, \quad (24)$$

provided the denominator is nonzero. For a positive tail formulation with $\tau = P(\rho) + y$, the denominator also equals $\ell^T \Psi_\xi(\rho, y)$, because the difference is $\ell^T (J - I)P'(\rho) = 0$. It is nonnegative, and is positive when some size-dependent monomial contributes positively.

Formula (24) suggests prioritizing verified defect improvements by their Perron-weighted value. It is conditional on the stated smooth critical family; it is not used in [theorem 4.1](#). In particular, it never authorizes subtracting an unproved or merely guessed defect.

9 Verification, formalization, and reproducibility

9.1 The proof dependency chain

The mathematical chain behind the new upper bound is

$$\begin{aligned} & \text{geometric recurrence lemma} + \text{exact marked prefix} \\ & \implies \text{nonnegative corrected tail system} \\ & \implies \text{rational supersolution at } 500/2249 \\ & \implies \text{coefficient majorant} \implies \lambda \leq 2249/500. \end{aligned}$$

The new lower certificate for the old equality system follows a different chain:

$$\begin{aligned} & \text{integer balance} + \text{rational log enclosure} \\ & \implies \text{no positive supersolution} \\ & \implies \mu_{\text{original}} \geq 452349/100000. \end{aligned}$$

These chains share the monomial map but not the logical direction of the majorant comparison.

9.2 Archive contents and commands

The archive includes the article source and PDF, the exact marked profile, twelve upper certificates, the dual certificate, the verification transcript, and source code. The principal files are:

File	Purpose
code/model.py	Sparse map, patterns, exact polynomial operations
code/verify.py	Standard-library rational certificate replay
code/enumerate.cpp	Incremental marked enumeration through size 18
code/enumerate_direct.cpp	Direct-scanning comparison implementation
code/research.py	Independent Python enumeration and numerical exploration
code/discover_certificates.py	Numerical search and rationalization
data/profiles.json	Exact marked and unmarked counts
data/upper_N18.json	Main rational upper certificate
data/dual_original.json	Balanced lower certificate for original model
data/verification.json	Exact arithmetic residuals and log interval

From the archive root, the fast arithmetic audit is

```
python3 code/verify.py --report data/verification.json
```

It uses only Python's standard library, arbitrary-precision integers, and `fractions.Fraction`. It checks the monomial graph hypotheses, nonnegative prefix defects, five occupancy partitions, all upper-certificate residuals and tail budgets, all dual balance equations, the rational log interval, and the repository's previous 4.5235 supersolution.

Full regeneration of the main finite table is separate:

```
c++ -O3 -std=c++17 code/enumerate.cpp -o enumerate
./enumerate 18 > regenerated.csv
python3 code/check_profile.py regenerated.csv
```

This visits more than two billion partial objects in total and is much slower than certificate replay. The comparison tool checks all marked and unmarked entries against the archived table. No external sequence database is needed for either enumeration or certificate checking.

The numerical discovery scripts require NumPy, SciPy, and SymPy. They are provided for further experimentation, but are outside the proof boundary. The figures in the numerical-estimate column of [table 3](#) should not be promoted to exact assertions merely because those scripts print a small residual or a successful solver status.

[write] *Layout in ProveIt*. The report directory keeps the delivered code/, data/ and tex/ layout, so the commands above run from the report root under the shipped names. Two files moved: the delivered Makefile is code/Makefile (its recipes still use root-relative paths, so run it as `make -f code/Makefile verify` from the report root), and requirements-discovery.txt is data/requirements-discovery.txt. The delivered PDF was replaced by a build of this text, and the delivered checksum manifest is not shipped. On the intake machine python3 is spelled py. Because `verify.py -report` rewrites data/verification.json, `make_tables.py` rewrites tex/*.tex, and `discover_certificates.py` overwrites data/upper_N*.json and data/dual_original.json, run them on a copy of the directory. At intake, `verify.py` passed on such a copy and its output equalled data/verification.txt.

9.3 A focused Lean extension plan

The most direct formal extension reuses `geometricPublishedBuiRecurrences` and the existing domination of the fixed-polyomino count by the G coordinate. The genuinely new interfaces are an exact finite prefix for the marked profile, the defect-corrected weighted-prefix inequality, and the new rational vector.

A practical route can avoid formalizing complex singularity theory entirely. First prove the finite convolution identity underlying (9). Next prove a weighted-tail version of the repository’s monotone-prefix certificate theorem. Instantiate it with the new vector, using ordinary exact rational reduction for every inequality. The existing growth endpoint can then carry the pointwise exponential bound to λ . The expensive part is a formally verified enumeration or a small proof object certifying the marked prefix, not the seventeen final inequalities. [write] In the Lean files at the pin, the “monotone-prefix certificate theorem” exists in ζ -generic form only as `buiIterate_le_supersolution` and `PrefixRecurrence.le_supersolution`; the endpoint theorems that carry a supersolution to a pointwise bound are specialized to $\zeta = 2000/9047$ (section 1.5). The final passage to λ can use the generic `growthSup_le_of_le_pow` (`Growth.lean:38`), since $v_G < 1$ gives $A_n \leq (2249/500)^n$.

For dual certificates, the balance identity is finite integer arithmetic. One can either formalize weighted arithmetic–geometric mean directly or expand rational weights into an unweighted finite product argument. A logarithmic checker with the remainder bound (21) avoids large powers and is a natural reusable component. The abstract completeness theorem need not be formalized before the particular 53-weight certificate can be checked.

No Lean source claiming these new results is included. In particular, the presence of a Lean proof of the old bound in the repository must not be confused with a Lean proof of the new finite-prefix theorem or of the size-18 data.

10 Proposed research questions and next projects

The following projects are intended as concrete continuations, not claims already established in this report.

1. **Kernel-check the 4.498 bound.** Formalize the finite-prefix certificate lemma and a verified marked polyomino enumerator, or design a compact reflective certificate for its output. The target should be an unconditional endpoint with precisely the same counting conventions as `GeometricComplete.lean`. This is the highest-priority validation project. [write] Section 1.5 lists the Lean pieces that exist, the ζ -specialization that must be generalized, and a smaller first target (the $N = 7$ certificate, $\lambda \leq 4.5233$, needs counts only through size 7).
2. **Generate longer marked prefixes without listing every object.** Adapt a transfer-matrix or connectivity-state enumeration to accumulate all seventeen occurrence counts, not merely A_n . The relevant challenge is preserving marked local information while merging equivalent boundary states. Exact counts of larger objects could strengthen the present bound without changing the imported geometric proof.
3. **Certify lower bounds for the corrected envelopes.** Expand or represent the positive maps Ψ_N and construct balanced-monomial certificates for them. This would replace the middle column of table 3 by narrow rigorous two-sided brackets. Sparsity or an arithmetic-circuit representation may avoid a large explicit expansion.
4. **Locate the actual stability barrier of the fixed Bui map.** Bound $\text{spr}(\Phi_X(\zeta, a(\zeta)))$ using certified lower and upper bounds on the true marked generating functions. Determine whether the prefix hierarchy’s limiting obstruction lies substantially below the true polyomino radius. The strict criteria in theorem 8.1 give concrete sufficient tests, while the equality case remains open.
5. **Optimize which geometric deficiencies to repair.** Use Perron-weighted sensitivities as a search heuristic, but accept a repair only after proving the corresponding injection or coefficient improvement. A small new neighborhood resolving a heavily weighted overcount may be more valuable than many additional unweighted states.

6. **Study Bui’s neighborhood-refinement convergence question.** Develop a hierarchy in which neighborhood size and geometric recurrences are refined together. Prove convergence to the true growth constant, or identify a rigorous obstruction. The scalar counterexample here rules out one naive prefix-only argument; it does not settle the neighborhood question.
7. **Quantify certificate bit complexity.** Given a rational target separated from the critical value by a known amount, bound the sizes of balanced integer weights and the number of logarithm terms required. Separate complexity of discovery from complexity of replay. The present million-unit total weight is an example, not a worst-case bound.
8. **Extend the alternative beyond the irreducible nonlinear case.** Decompose reducible systems into strongly connected blocks and analyze linear, unproductive, and boundary blocks separately. A comprehensive theorem should explain when weak infeasibility can occur and when a strict rational obstruction still exists. Maximum operators require a different or enriched certificate language.
9. **Obtain certified critical asymptotics.** For selected corrected systems, prove uniqueness of the relevant positive critical point, nondegeneracy, and the resulting coefficient asymptotics. Such a result would justify numerical critical continuation as mathematics, not merely as a source of candidate certificates.
10. **Transfer the method to other lattice animals and local grammars.** Identify an existing positive recurrence system on a different lattice, compute its first exact defects, and test whether a modest prefix correction produces a useful improvement. The general theorems apply without change once properness and the underlying combinatorial domination are proved.

11 Conclusion

The repository’s polyomino project supports more than a small numerical retuning. Incorporating exact marked prefixes changes the majorant itself, and the resulting rational certificate proves $\lambda \leq 4.498$. Balanced monomial weights provide the missing lower-certificate mechanism for a natural class of convolution systems; their explicit use also proves that the original recurrence cannot support the new bound without a structural correction. Finally, the spectral analysis identifies why this successful improvement must not be extrapolated automatically to asymptotic exactness. The central next step is independent verification, especially formal certification of the finite marked enumeration.

A Complete marked profiles through size eighteen

The tables give the exact occurrence counts for the patterns in [table 1](#). All omitted size-zero entries are zero. The row A_n counts translation classes; every other row counts marked occurrences. These data, together with (6), determine every coefficient of D_{18} by finite integer convolution. The JSON and CSV forms contain the same integers without typographical formatting.

n	1	2	3	4	5	6	7	8	9
A_n	1	2	6	19	63	216	760	2725	9910
C	1	1	1	3	10	35	128	479	1819
D	1	1	2	6	20	71	260	973	3696
E	1	1	3	10	35	128	479	1821	7001
F	1	3	10	35	128	481	1841	7131	27856
G	1	2	6	20	71	261	982	3751	14481
H	1	2	5	15	51	182	670	2517	9592
P	0	1	4	15	57	220	859	3380	13375
Q	0	1	3	10	36	133	503	1930	7480
R	0	1	3	8	25	85	303	1110	4150
S	0	1	3	9	31	111	410	1544	5896
T	0	1	3	9	30	106	388	1451	5510
U	0	1	4	14	50	185	700	2686	10412
V	0	0	1	4	14	52	196	748	2884
W	0	0	1	4	13	46	168	625	2362
X	0	1	2	5	16	54	192	703	2626
Y	0	1	2	4	12	39	135	485	1788
Z	0	1	2	3	8	25	84	296	1080

Table 5: Exact profiles for sizes 1 through 9.

n	10	11	12	13	14	15	16	17	18
A_n	36446	135268	505861	1903890	7204874	27394666	104592937	400795844	1540820542
C	6979	26988	105011	410629	1612207	6351215	25091789	99372469	394381157
D	14190	54917	213865	837024	3289287	12969734	51285522	203287656	807484165
E	27144	105927	415489	1636445	6467194	25630675	101822518	405337049	1616425239
F	109503	432550	1715129	6821554	27199077	108673645	434959946	1743471996	6997283445
G	56348	220593	867716	3426361	13572506	53905206	214569185	855714279	3418228578
H	36929	143269	559155	2192740	8632427	34094245	135023622	535974728	2131792166
P	53155	211957	847413	3395193	13626571	54768439	220390761	887757717	3579054867
Q	29204	114666	452227	1789916	7105312	28274531	112746667	450377230	1801803339
R	15744	60366	233323	907529	3547802	13926810	54857224	216706457	858185665
S	22739	88352	345290	1355716	5343140	21124511	83738100	332687072	1324308001
T	21162	81966	319543	1252166	4927255	19455297	77039780	305805462	1216400095
U	40678	159874	631249	2501630	9943629	39622046	158205688	632793092	2534808294
V	11212	43866	172467	680772	2695907	10704713	42601646	169868638	678451341
W	9035	34875	135550	529733	2079362	8191718	32369106	128231694	509107371
X	9950	38100	147076	571394	2231348	8750584	34438134	135936824	537948754
Y	6709	25491	97773	377796	1468440	5735092	22488118	88474763	349078294
Z	4025	15214	58108	223734	866978	3377002	13210216	51861498	204221945

Table 6: Exact profiles for sizes 10 through 18.

B A minimal exact certificate checker

The following core is sufficient to check an upper candidate once the map and true prefix are supplied. The full verifier also checks the input conventions, all intermediate certificates, the dual weights, and its rational logarithm enclosure.

```

from fractions import Fraction as Q

# P[i] and D[i] are integer coefficient lists, low degree first.
def evaluate(coefficients, zeta):
    value = Q(0)
    for coefficient in reversed(coefficients):
        value = value * zeta + coefficient
    return value

def check_upper(zeta, values, P, D, polynomial_map):
    prefix = [evaluate(p, zeta) for p in P]
    defect = [evaluate(d, zeta) for d in D]
    image = polynomial_map(zeta, values)
    tail = [v - p for v, p in zip(values, prefix)]
    residual = [v - f + d
                 for v, f, d in zip(values, image, defect)]
    if any(t < 0 for t in tail):
        raise ValueError("Candidate lies below the known prefix")
    if any(r < 0 for r in residual):
        raise ValueError("A supersolution inequality failed")
    return tail, residual

```

The defect is computed, not guessed. If $P_i(\xi) = \sum_{n=1}^N a_{i,n} \xi^n$, then for each row i one expands the finitely many monomials in $\Phi_i(\xi, P)$, discards degrees above N , and subtracts P_i . Because all numbers involved are integers before evaluation, this step is particularly suitable for independent implementations or proof-assistant reflection.

For the dual checker, (16) is a sum of integer vectors. The j th entry of a monomial exponent vector is counted with its full multiplicity. Omitting the factor two for a square, for example, would invalidate the certificate. The supplied verifier iterates over repeated variable indices, so this multiplicity is explicit in its sparse format.

C Source audit and claim classification

Claim or ingredient	Status and evidence
Geometric inequalities	Imported from Bui v2, Lemma 4 and Appendix B; corresponding repository endpoint source inspected at the pinned commit. [write] In ProveIt a Lean theorem for the actual counts, <code>geometricPublishedBuiRecurrences</code> (section 1.5).
Old 4.5235 upper bound	Existing repository result; rational vector independently replayed in the new verifier. No new Lean build was performed. [write] Kernel-checked in ProveIt; the project's Lean-verified bound.
Prefix-corrected majorant	Proved in section 3, including the nonnegative tail map and the known-prefix lower-budget condition.
New 4.498 bound	Computer-assisted theorem: exact marked enumeration plus a rational certificate, with the detailed dependency chain in section 9.

Claim or ingredient	Status and evidence
Marked table through 18	Generated by exhaustive C++ enumeration. Python agrees through 11; direct and incremental C++ scans agree through 16; standard unmarked totals agree through 18. Not kernel-checked. [write] Regenerated through 18 at intake from the final source, and through 10 by an independent enumerator (section D).
[write] New 4.498 bound in Lean	Not formalized; the Lean endpoints are specialized to $\zeta = 2000/9047$ (section 1.5).
Dual obstruction	Elementary weighted AM–GM proof; particular integer balance and strict log inequality replayed exactly.
Certificate completeness	Proved for productive, strongly connected, nonlinear positive polynomial systems with rational data. Not claimed for all maximum or signed recurrences. Based on classical geometric-programming ideas.
Numerical critical values	Exploratory estimates only; not certified optimal values and not used to establish the accepted inequalities.
Stability theorem and gap	Written proofs in section 8 . The case of limiting spectral radius exactly one is not classified generally.
Sensitivity formula	Conditional on an already justified differentiable critical family. Not used for the headline bound.
Novelty	The explicit bound and application were not found in the inspected sources. No claim of an exhaustive priority search or independent refereeing.

D [write] Provenance, intake verification and non-claims

D.1 Provenance

This report was built from one manuscript: manuscript 06 of ProveIt’s incoming-reports batch 36, the archive ProveIt_Polyomino_Research (delivered in commit e13affd32, placed in commit 1a1396d4d). It was written against the pin e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9; the project it continues is unchanged between that pin and the placement commit, so its statements about the project are current except where the delivered text undersells the formal status of the geometric lemma ([section 1.5](#)). As a single source it required no merge choices. Its whole text is printed; passages marked [write] were added when it was written into ProveIt: [sections 1.4](#) and [1.5](#), this appendix, the paragraph on earlier upper bounds in [section 1](#), notes in the title-page box and the “three levels” box, after (8) and the geometric-lemma paragraph of [section 2](#), on intake regeneration in [section 4](#), on layout and on the Lean plan in [section 9](#), after research question 1, three rows of the audit table in [section C](#), and four bibliography entries ([7, 8, 9, 10]). All labels carry the prefix kfp: except eq:map, which the delivered, generated tex/dual_table.tex cites by that name.

D.2 Intake verification

- The arithmetic verifier code/verify.py passed on a copy of the package (4.3 s) and printed exactly data/verification.txt; the profile checker accepted data/profiles_18.csv.
- The size-18 table was regenerated from scratch by a fresh build of code/enumerate.cpp and matched in every entry ([section 4](#)); fresh runs through sizes 11–14, 16 and 17 and a direct-scan run through 12 matched too.

- An independent set-based Python enumerator, with coordinates from the project's `Patterns.lean`, matched all entries through size 10. An independent transcription of (6) from the project's research note [7] reproduced (13), the size-18 margins and v_G , and confirmed the old 4.5235 vector as a supersolution.
- The seventeen offset sets of table 1 were compared entry by entry with `Patterns.lean`, and the unmarked column with OEIS A001168 through 18.
- A 50-digit evaluation of $\log C(\zeta_{\text{dual}}, m)$ gave 1.00066802314520877 . . . , inside (20).

These checks do not verify the written proofs, and no independent proof review was performed. At intake the proofs of lemma 3.2, theorems 3.3, 3.4, 3.6, 5.1, 6.3 and 8.2 were checked and that of theorem 8.1 was read; no gap was found.

D.3 What this report does not claim

Every limitation stated by the delivered text is kept; collected:

- N1. The lower bound $\mu_{\text{original}} \geq 452349/100000$ concerns the uncorrected equality recurrence and is *not* a lower bound on λ (section 5).
- N2. No new Lean formalization. The Lean proof of 4.5235 in ProveIt is not a proof of 4.498, of the finite-prefix theorem, or of the size-18 data.
- N3. The arithmetic checker does not prove that the table enumerates the stated objects; its partition, defect and G -domination checks are necessary conditions only.
- N4. The enumeration cross-checks are not a formal certificate that every size-18 object was generated exactly once. The delivered record made no fresh size-18 run after the final command-line changes; the intake regeneration closes that gap only with the same program, and the two C++ implementations share one generation strategy.
- N5. The report is an AI-assisted draft and has not been independently refereed.
- N6. Novelty is assessed only against the inspected repository snapshot and sources; there was no exhaustive priority search.
- N7. Geometric programming, weighted AM–GM and Redelmeier-style enumeration are prior tools, not claimed inventions.
- N8. The certificate alternative is proved only for productive, strongly connected, nonlinear systems with rational data; it gives no uniform bound on certificate bit length and does not cover maximum or signed recurrences.
- N9. The case $\text{spr}(J) = 1$ of theorem 8.1 is not classified.
- N10. The counterexample of theorem 8.2 does not disprove Bui's Question 1.
- N11. The sensitivity formula (24) is conditional and not used in theorem 4.1.
- N12. The numerical critical estimates in table 3 are not certified values or bounds.
- N13. The OEIS comparison is a consistency check, not an input to the proof or the checker.

Added in the write: (W1) the geometric lemma is kernel-checked in ProveIt, but only relative to the transcription of Bui's diagrams into `Patterns.lean`; (W2) the bound 4.498 remains unverified by Lean, and the project's Lean-verified bound remains 9047/2000; (W3) the earlier bounds of section 1 were not re-examined, and their sources were not read beyond their bibliographic data.

References

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